

SYSTEMS ENGINEERING

Systems Analysis & Design

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2025-I



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Basic Concepts |
- 2 Human Activities |
- 3 Cybernetics and Technology |
- 4 Teams-Based Structure as a System |



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- 1 Basic Concepts
- 2 Human Activities
- 3 Cibernetics and Technology
- 4 Teams-Based Structure as a System



What is Systems Engineering?

- **Systems Engineering** is a **discipline** that studies the **design**, **implementation**, and **maintenance** of **complex systems**.
- This **discipline** is based on interdisciplinary fields, such as control engineering, industrial engineering, software engineering, mechanical engineering, electrical engineering, organizational studies, project management, and others.
- **Systems Engineering** is a holistic approach to engineering that focuses on how to **design** and manage complex systems over their lifecycle.



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Requirements -> Analysis & Design -> Code -> Test ->
Deploy



Systems Engineer as a Professional

- A **Systems Engineer** is a professional who is responsible for designing, implementing, and maintaining complex systems.
- A Systems Engineer must have a broad understanding of engineering, mathematics, science, and technology.
- A Systems Engineer must be able to analyze and solve problems in a systematic and holistic way.



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Xerox



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Human Activities

- Human activities are complex systems that involve multiple components, interactions, and feedback loops.

- Systems Engineering can be applied to understand, analyze, and improve human activities.

- Systems Engineering can help us design and manage complex systems such as organizations, cities, economies, and societies.

Game Of Life



live - 3 → live
 live < 3 → dead
 > 3 → dead
 dead → 3 → live



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7 1000 0.1
7:05 0. 0.3

Algorithmic
Trading



Human Activities: Complexity

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- **Interactions** are key to understanding and improving human activities.
- **Organizations** are complex systems that involve multiple departments, teams, and individuals.



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Organizations as Systems

- **Organizations** can be **viewed** as: a **rational system**, a **natural system**, or an **open system**.
- A **rational system** is a formal organization that is designed to achieve specific goals as a machine.
- A **natural system** is an informal organization that is emergent and adaptive based on human interactions.
- An **open system** is an organization that is interconnected with its environment and adapts to changes.



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ψ - 3000



Sinergy

- **Sinergy** is a simple but powerful concept: the aim of the parts is more than the parts itself.

- It means the **interactions** could boost the capabilities of the parts of the system. Also, it lets both understand **emergent behaviors** and define improvements in **systems**.

- One of the main concepts is the **theory of the computation**. Based on graphs, you could define a **computational machine**.

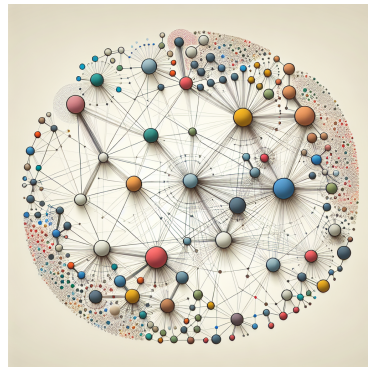


Figure: Prompt: Define a draw of clusters in social networks.



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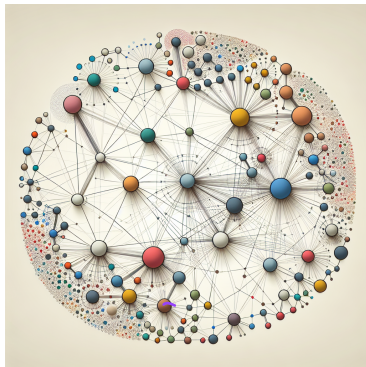


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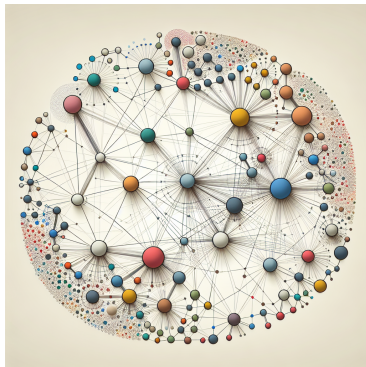
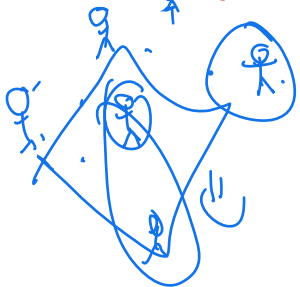


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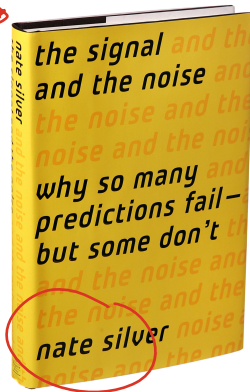


Synergy: Money Ball

Athletics \$ 15

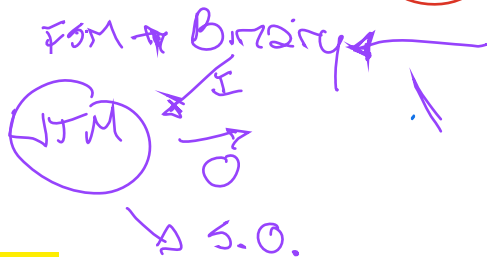


The Outliers



Talking with Machines!

Alan Turing (1912-1954)
 ↳ Babbage Machine
 ↳ State Machine



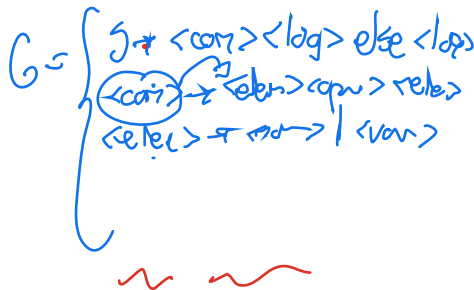
- Alan Turing proposed a hundred years ago an Universal Machine, capable of take any algorithm defined as a state machine, and process it in a binary language.

Thesis Turing-Chomsky
 Chomsky proposed the formal
 Algorithm based on generative
 grammars. Here is where the
 high-level programming
 languages appear.

FOM
 ↓
 Machine



Talking with Machines!



- **Alan Turing** proposed a hundred years ago an **Universal Machine**, capable of take any algorithm defined as a **state machine**, and process it in a **binary language**.
- Forty years after, **Noam Chomsky** proposed the **formal languages** based on **generative grammars**. Here is were the **high-level programming languages** appear.



Programming Languages

- **Programming Languages** with more capabilities, easier comprehension had been created. Also, more people start to code into specific domain programming languages.

- **Andrej Karpaty**, hero in Tesla Company and now in OpenAI said: *nowadays, english is the more important programming language.*

8-17

C++ / Java

$x > 5$? true: false;

Python

true if $x > 5$ else false

↓

natural language

>>> 0)



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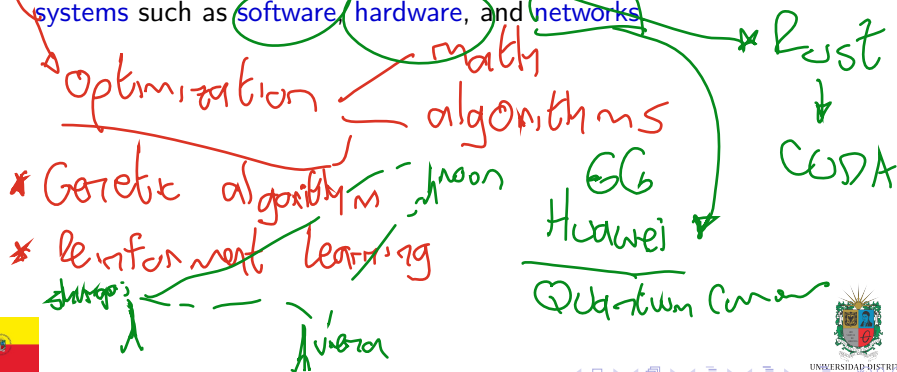
2023
! copilot → codeX



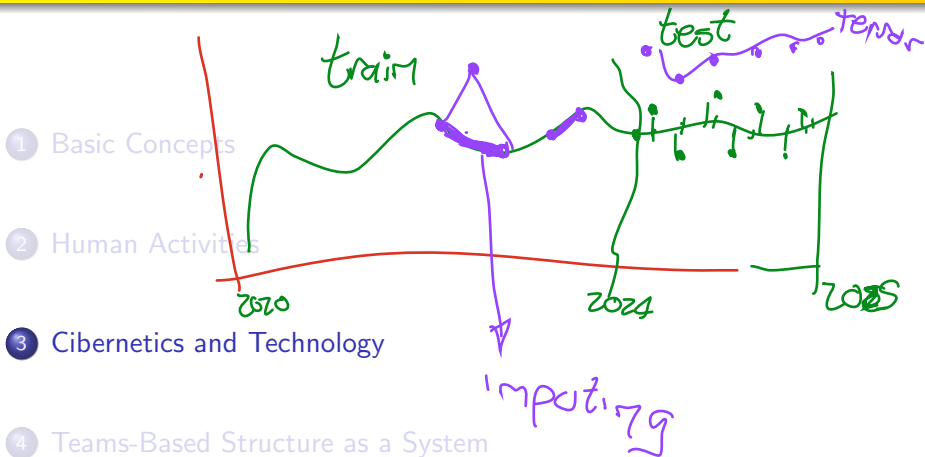
Applications of Systems Engineering

Qubits

- **Systems Engineering** can be applied to understand, analyze, and improve complex systems.
- **Systems Engineering** can be applied to design and manage complex systems such as organizations, cities, economies, and societies.
- **Systems Engineering** can be applied to design and manage complex systems such as software, hardware, and networks.



Outline



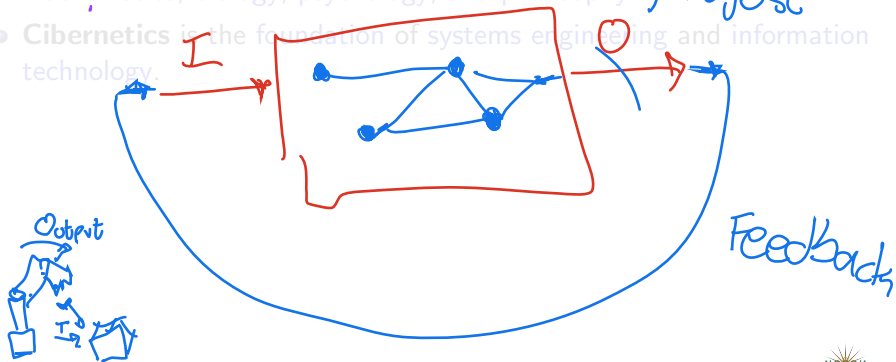
Cibernetics → Robots

- **Cibernetics** is the study of systems, control, and communication in animals, machines, and organizations.

Environment

- Cibernetics is a transdisciplinary field that combines engineering, mathematics, biology, psychology, and philosophy.

- Cibernetics is the foundation of systems engineering and information technology.

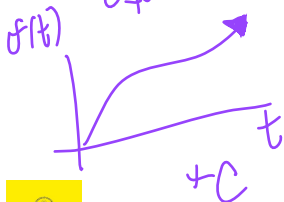


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derives
differential
equations



bio-inspirations
robots

Control
mechanic
electronic
system

kid
less
errors

learning

Yam le
Cam
Zama



Cibernetics

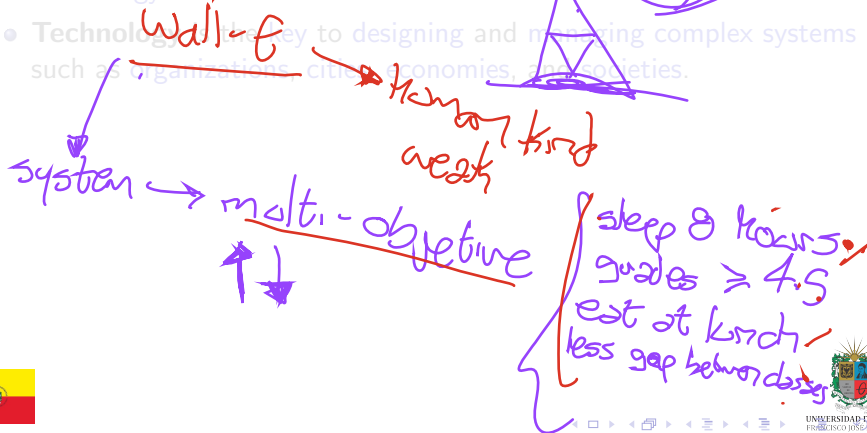
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Charlie Sheen → environment



- Technology is the key to designing and managing complex systems such as organizations, cities, economies, and societies.



Technology

- **Technology** is the application of scientific knowledge to solve problems and improve systems.
- **Technology** is the foundation of systems engineering and information technology.

Technology is the key to designing and managing complex systems such as organizations, cities, economies, and societies.

Handwritten notes:

- cell phones
- internet
- software development role
- hardware fixer



Technology

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Speed
Communication

Sensors

IPots

Quality



Technology: AI

Agents

- **Artificial Intelligence** is a field of computer science that studies how to design and implement intelligent agents.

- Artificial Intelligence is the key to designing and managing complex systems such as smart organizations, smart-cities, smart-economies, and smart-societies.



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schedule
machines
flow stop
teams → skill matrix

Algorithmic Training

2009 (The Big Chart)

2007

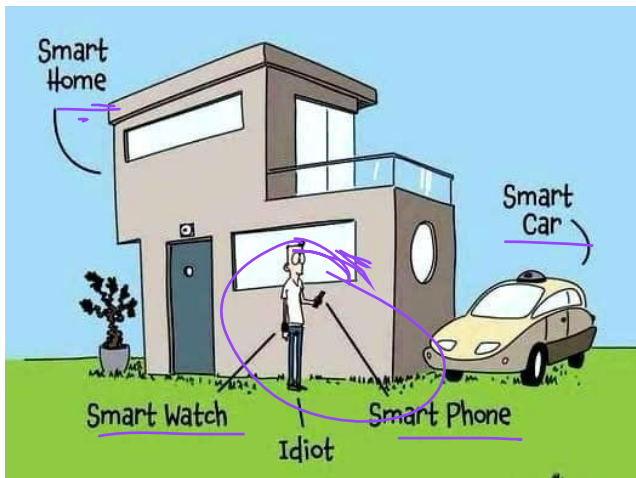
Big

roadway
lights smart

Im



Artificial Intelligence **not** as a System



Cibernetics & Technology in Systems Context

Feedback

- **Cibernetics** and **technology** are the foundation of **systems engineering** and **information technology**.
- **Cibernetics** and **technology** are the foundation of **artificial intelligence** and **smart systems**.

34

Smart house



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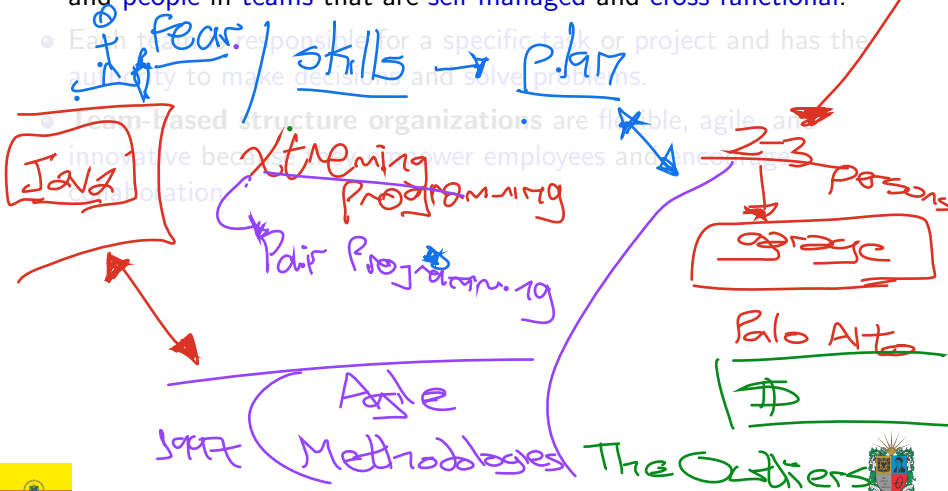


Team-Based Structure Organizations

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- Each team is **responsible** for a specific task or project and has the **authority** to make decisions and solve problems.

- **Team-based structure organizations** are **flexible**, **agile**, and **innovative** because they **empower employees** and **encourage collaboration**.



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- Two → Atom House*
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Ownership

Junior

Hired-Selection Process

- ① Call Psychology - Recruiter / RHH → *English*
- ② i) Call Lead Tech ii) Technical Test

Contest

$$abs(x_3 - x_4) == abs(y_2 - y_1)$$

hackerrank
leetcode

Moving Average

WOD

*Globalt
me
Graz*

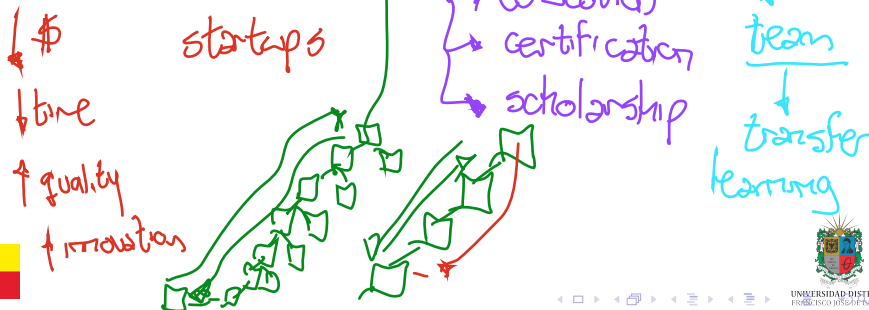
*soft
skills*

Fury



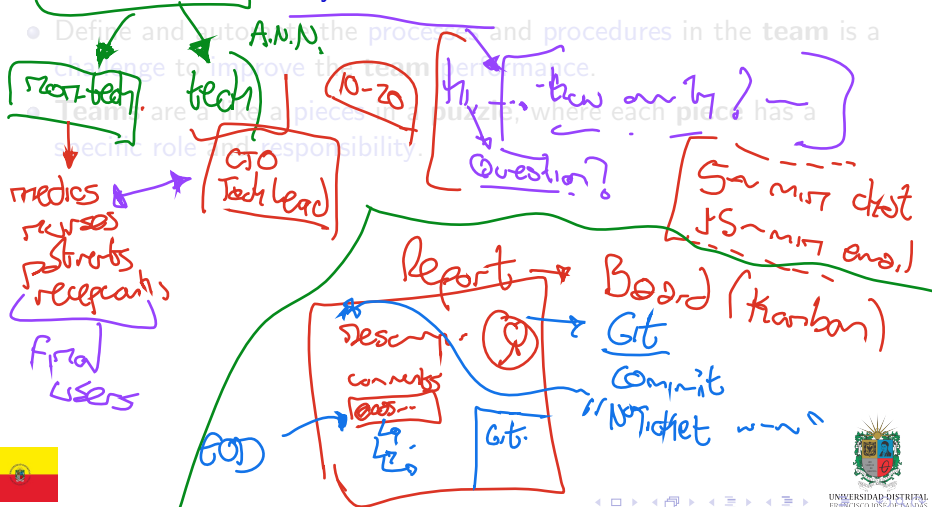
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- Define and automate the processes and procedures in the team is a challenge to improve the team performance.

- Teams are like a pieces in a puzzle, where each piece has a specific role and responsibility.



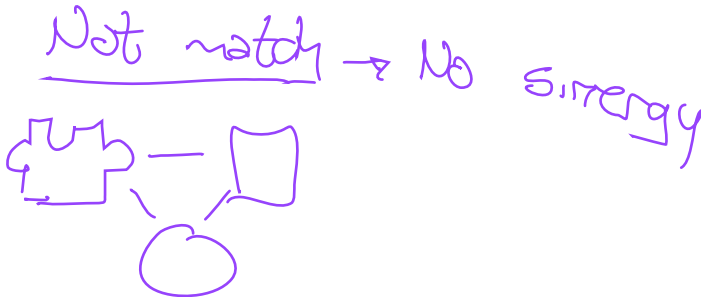
Teams as a System

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Soft Skills

hard



- **Soft skills** are **personal attributes** that enable someone to **interact** effectively and **harmoniously** with other **people**.

Typical Soft Skills:

- **Communication** skills (verbal and written).
- **Teamwork** and **collaboration**.
- **Problem-solving** and **critical thinking**.
- **Adaptability** and **flexibility**.
- **Time management** and **organization**.
- **Leadership** and **management**.
- **Emotional intelligence**.
- **Creativity** and **innovation**.
- **Conflict resolution**.
- **Networking** and **relationship building**.
- **Customer service** and **client management**.

Tools

team



Variables



Computer Analyst

Skills:

- Business process modeling and documentation.
- Data analysis and interpretation.
- Requirements gathering and management.
- Stakeholder management.

Responsibilities:

- Analyzing business processes and identifying areas for improvement.
- Gathering and documenting business requirements.
- Collaborating with stakeholders to define project scope and objectives.
- Creating and maintaining project documentation, including functional specifications and use cases.
- Facilitating communication between business users and technical teams.
- Participating in system testing and user acceptance testing.
- Providing support and training to end users.

Statistics → Boyes → ML

Descriptive Analysis

Prescriptive Analysis

Activity 4 Sequence

Dashboard Client

Bizagi

B



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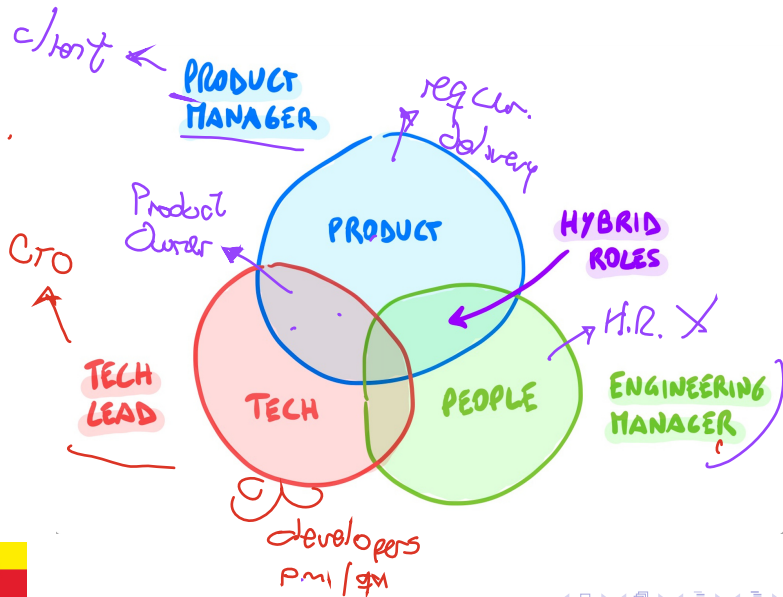
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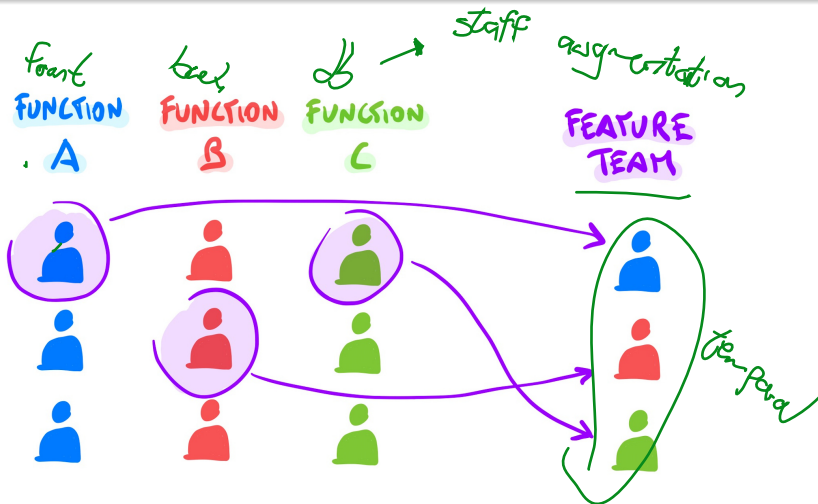
Expectations



Tech Company Typical Structure



Feature Teams



What is to be a leader? I

- Leading a team is not a role. It is a decision, you could be a leader anytime and anywhere.
- Teamwork culture is pretty important. It creates habits, open communication, safety, spaces for inclusion.
need? → why?
- Psychological safety is a key point to have an effective team. You could develop technical skills, but *can't se sources?*
- Hierarchy is very important. Anarchism tends to fail. Hierarchy exists by status and power.
- In a hierarchy experts lead to make better decisions. However, anyone must be careful to not leave people behind.



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right? → communication?
channel?



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2 days
Python



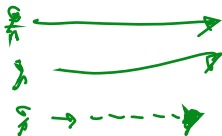
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1:1 ~ week
bi-weekly
15 ~ 30 min



What is to be a leader? II

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- A **good leader** should focus on **outcomes** rather than **outputs**. This helps bring **business value** rather than **just complete tasks**.
- **Failure** is always an **option**. Learn how to **deal** with **setbacks**; do not punish — just **fix and learn**.
- Some believe you're **born a leader**, while others think that a leader can be **developed** over time. Either way, **context** and **the desire** for self-growth are vital.
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process

improvement



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The Outliers



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What is to be a leader? III

hard

- It is important to maintain **psychological well-being**. You will be stronger, better able to help people, and have a better perspective on everything.
- A **good leader** builds **trust relationships** and has the **emotional intelligence** to communicate effectively and understand others.
- To develop as a leader some good **guidelines** are the **three C's**: Curiosity, Courage, and Commitment.



What is to be a leader? III

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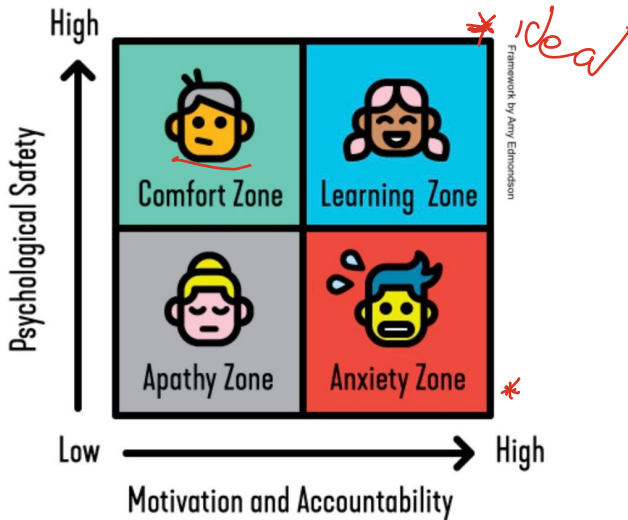


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Working Zones



Outline

- 1 Basic Concepts
- 2 Human Activities
- 3 Cibernetics and Technology
- 4 Teams-Based Structure as a System



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

